

IAS PRELIMS/MAINS

Ecology & Environment

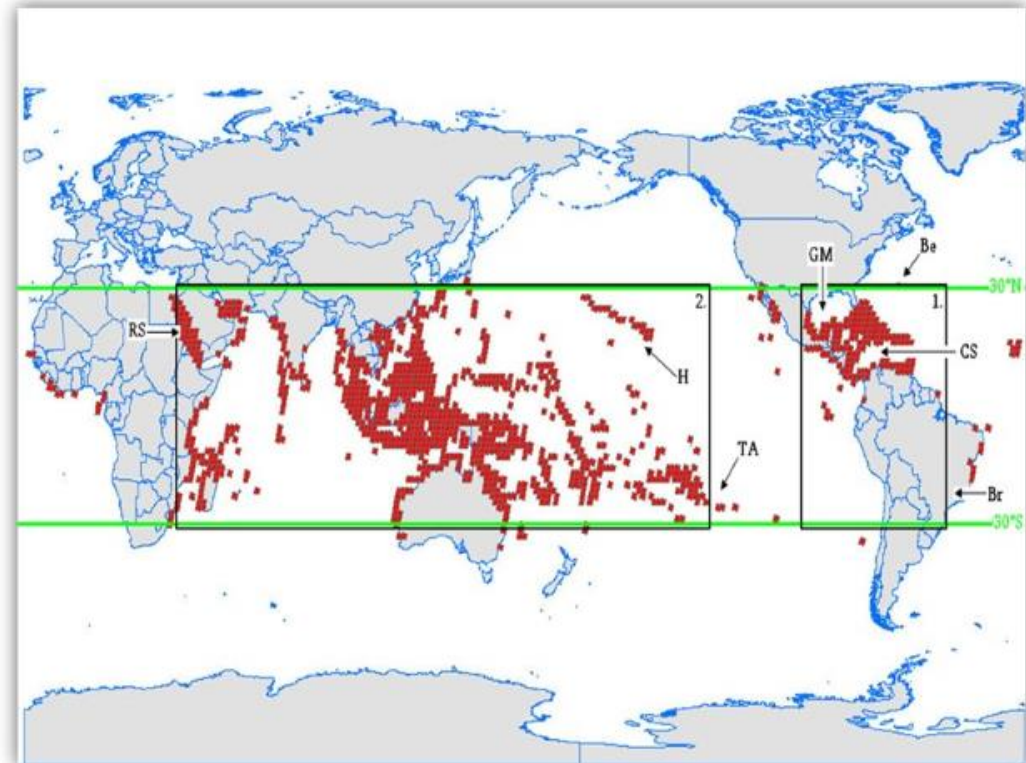
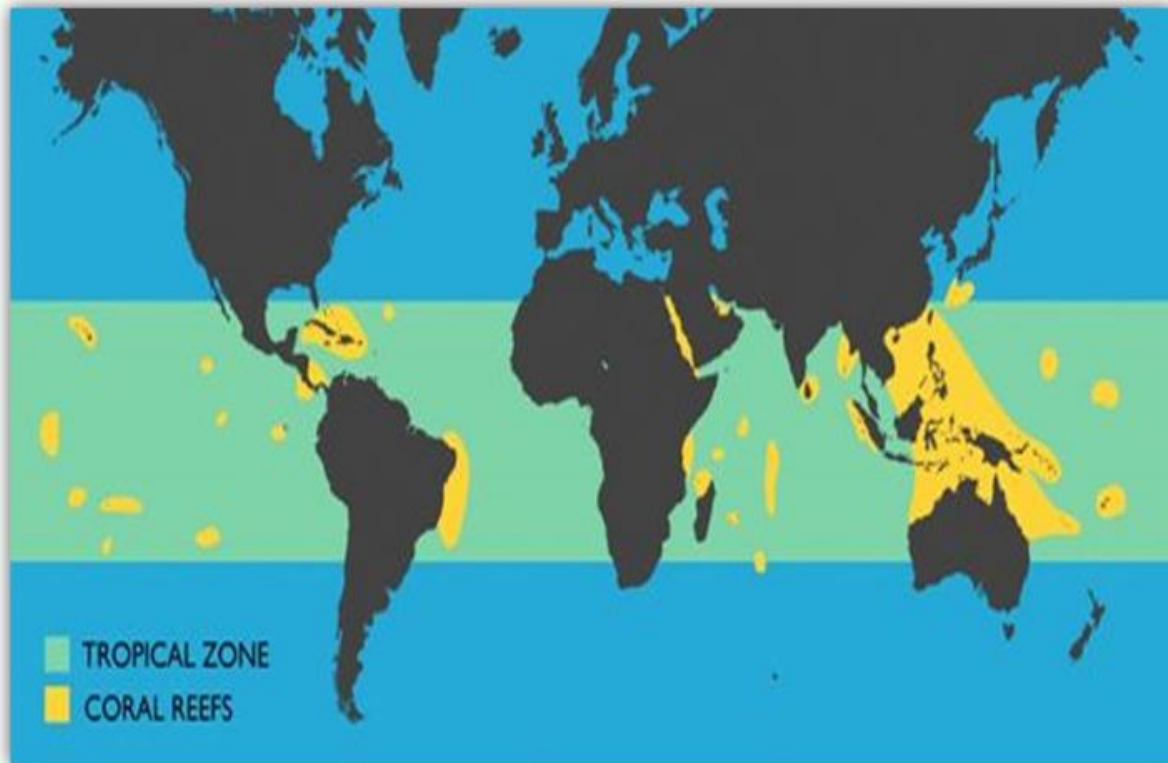
A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 27

WATER POLLUTION - 4

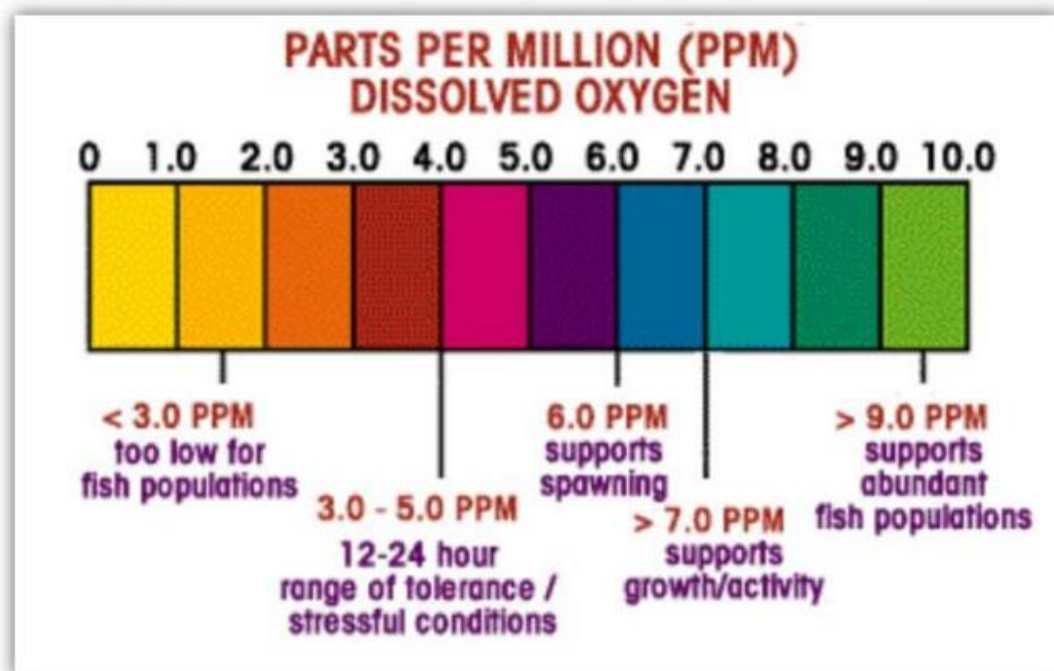
DISTRIBUTION OF CORAL REEFS



WATER QUALITY PARAMETERS

1 DISSOLVED OXYGEN (DO)

- **Definition** - The amount of oxygen present dissolved in water. It is measured in terms of milligrams per liter (mg/liter).
- Presence of organic and inorganic wastes in the water decreases the Dissolved Oxygen (DO) content in the water.
- Water having Dissolved Oxygen content less than 8 mg/l may be considered as contaminated and water having Dissolved Oxygen content below 4 mg/l is considered to be highly polluted.
- The higher amounts of organic wastes present in the water increase the rate of decomposition by bacteria and dissolved oxygen consumption, thereby decreasing the oxygen content of the water.



WATER QUALITY PARAMETERS

2

BIOCHEMICAL/BIOLOGICAL OXYGEN DEMAND (BOD)

- Water pollution by organic waste is measured in terms of biochemical oxygen demand.
- **Definition** - Biochemical oxygen demand is the amount of dissolved oxygen needed by the bacteria to decompose the organic matter present in water.
- Higher value of biochemical oxygen demand indicates low Dissolved Oxygen content of water.
- Biochemical Oxygen Demand can only be used for measuring organic waste in water. It is measured in terms of mg/liter.



WATER QUALITY PARAMETERS

3 CHEMICAL OXYGEN DEMAND (COD)

- Chemical Oxygen Demand is the amount of Oxygen in Parts Per Million (PPM) required to Oxidize the organic and oxidizable inorganic compounds present in the water.
- COD shows an even accurate picture of Water Quality compared to BOD.



VARIOUS SOURCES OF WATER POLLUTION

- 1. Domestic sewage** - It contains harmful micro organisms and nutrient rich content (Nitrates and Phosphates) which causes Eutrophication.
- 2. Industrial Discharge** - The industries discharge several inorganic and organic pollutants, which are highly toxic to the living beings.
 - ✓ Discharge of waste water from industries like petroleum, paper manufacturing, metal extraction and processing, chemical manufacturing, etc., often contain toxic substances like, heavy metals such as mercury, cadmium, copper, lead, arsenic and a variety of organic compounds.



VARIOUS SOURCES OF WATER POLLUTION

3. **Pharmaceutical Industries** – Organic solvents and other harmful chemicals.
4. **Agricultural sources** - Agricultural runoff- Phosphates, Nitrates, Ammonia etc. Pesticides and insecticides –Chlorinated Hydro carbons like DDT, Endosulfan etc.

ENDOSULFAN

- ✓ Endosulfan is an insecticide. In addition to its application on food crops, Endosulfan can be used as a wood preservative as well.
- ✓ Endosulfan is gradually being phased out on a global scale due to the wide range of adverse health effects. It is a highly potent neurotoxin.
- ✓ The health effects include – children being born with disabilities, affecting endocrine glands, causing toxicity and having the potential to bio-accumulate. (Endosulfan is a POP)
- ✓ All of these factors have led to more than 50 countries banning Endosulfan. (Stockholm Convention on POPs)
- ✓ In India despite laws banning it, it is still used in some places.



THERMAL AND RADIATION POLLUTION

- Thermal and nuclear power plants use a lot of water for cooling purposes and then discharge the hot water into a nearby water body.
- Discharge of such extremely hot water increases the temperature of the water body by 10 or 15D C above the ambient water temperature, this is called Thermal Pollution.
- Increase in water temperature decreases DO level in the water.
- It also affects the metabolism, growth and swimming of aquatic organisms.
- Nuclear accidents near water bodies pose major risk of radiation leakage into the water.
- Radioactive Iodine absorbed by thyroid gland causes thyroid cancer.



- An oil spill is the release of a liquid crude oil or petroleum into the environment, especially the marine ecosystem due to human activity.
- The term is generally used for marine oil spills, where oil is released into the ocean or coastal waters, but spills may also occur on land.
- Oil spills at sea are generally much more damaging than those on land, since they can spread very fast for hundreds of nautical miles in a thin oil slick (a thin film of oil) that covers the water.
- Oil being lighter than water covers the water surface as a thin film cutting off Oxygen to phytoplankton and aquatic plants and animals.
- Within a few hours of an oil spill, the aquatic organisms die due to suffocation and metabolic disorders.
- Birds and animals that consume dead fishes and plankton also die due to poisoning.
- Oil spill impacts beach tourism and fishing along the coasts.



INVASIVE SPECIES

- Aquatic plants like Water Hyacinth also called the Terror of Bengal are one of the most problematic invasive aquatic weed species.
- They grow abundantly in water bodies, especially in those water bodies which are nutrient rich.
- They cause havoc by their excessive growth which causes imbalance in the ecosystem. They also lead to stagnation of water in rivers.
- They destroy the ecosystem by causing imbalance and killing other organisms by depleting the resources from the ecosystem.



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